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Equilibrium Shape of Colloidal Crystals

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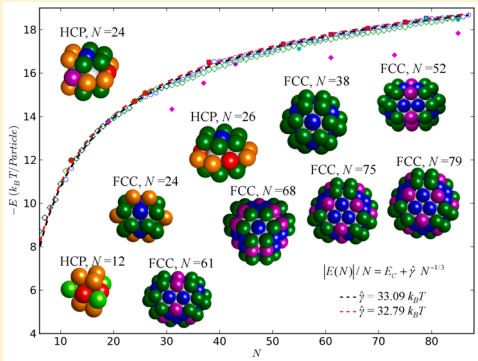
ABSTRACT: Assembling colloidal particles into highly ordered configurations, such as photonic crystals, has significant potential for enabling a broad range of new technologies. Facilitating the nucleation of colloidal crystals and developing successful crystal growth strategies require a fundamental understanding of the equilibrium structure and morphology of small colloidal assemblies. Here, we report the results of a novel computational approach to determine the equilibrium shape of assemblies of colloidal particles that interact via an experimentally validated pair potential. While the well-known Wulff construction can accurately capture the equilibrium shape of large colloidal assemblies, containing $O(10^4)$ or more particles, determining the equilibrium shape of small colloidal assemblies of $O(10)$ particles requires a generalized Wulff construction technique which we have developed for a proper description of equilibrium structure and morphology of small crystals. We identify and characterize fully several "magic" clusters which are significantly more stable than other similarly sized clusters.



1. INTRODUCTION

The assembly of colloidal particles into highly ordered configurations has significant potential as a convenient and efficient fabrication route toward enabling a wide range of technological applications. Of particular interest is the development of materials, such as photonic crystals, which exhibit a photonic band gap; photonic band gap engineering will allow for applications in analogy with those based on electronic band gap engineering in semiconductor materials.[1−3] Such engineering advances require fundamental advances in our understanding of the structure formation and growth of colloidal assemblies to match our deep fundamental understanding of atomic structure formation and growth of semiconductor crystals. While the thermodynamic bulk-scale crystalline structure has been well characterized for colloidal systems,[4,5] the structure of large, yet finite colloidal assemblies, containing $O(10^4)$ particles, has not been fully characterized. Additionally, small colloidal assemblies, which are even further removed from the thermodynamic bulk scale, e.g., assemblies containing $O(10)$ particles, have been shown to exhibit significantly different behavior compared to the respective bulk-scale phases.[6−9] These small clusters of colloidal particles are of particular interest not only because they play a significant role in the formation of stable crystalline nuclei that dictate the growth of colloidal crystals but also because they may constitute the building blocks for new classes of photonic crystals. For example, if we can identify clusters that exhibit enhanced stability compared to similarly sized clusters, i.e., clusters that are often referred to as "magic" clusters, we may use these stable clusters as individual unit constituents in a crystalline assembly (much like individual atoms in atomic crystals) with properties that can be tuned by controlling both the size of these magic clusters and the crystalline (lattice) structure in which they are arranged.

An important first step in describing the thermodynamics and kinetics of a crystalline assembly is to construct the minimum-energy configurations of the assembly as a function of its number of interacting particles.[10] One approach to determining the minimum-energy configuration of colloidal assemblies is to apply the well-known Wulff construction.[11,12] While the Wulff construction is used routinely for determining the equilibrium shape of atomic crystals, such as metals and semiconductors, such a study has not been undertaken for colloidal crystals. However, it is unclear whether the governing assumptions which underpin the Wulff construction are valid for the description of all the relevant energetic effects in a crystalline assembly of colloidal particles.[13]

Here, we present the findings of a computational study for the equilibrium-shape, minimum-energy configurations for crystalline clusters of colloidal particles which interact via an experimentally justified pair potential described in the simulation and analysis methods below. These minimum-energy configurations are constructed in two ways: first, by the application of the well-known Wulff construction,[14] which is successful for systems that are near bulk in size, i.e., assemblies of $O(10^4)$ particles or larger, and, second, by developing and implementing a generalization of the Wulff construction to successfully describe the equilibrium state of small systems, which contain $O(10)$ particles. This generalization of the Wulff construction is capable of determining the minimum-energy crystalline configurations for given choice of crystalline packing,

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such as face-centered cubic (FCC), hexagonally close-packed (HCP), body-centered cubic (BCC), or any other lattice structure. The results of the generalized Wulff construction are analyzed to identify so-called "magic" clusters, which exhibit improved stability compared to that of similarly sized clusters. We also describe the morphology of these clusters using a novel core−shell characterization approach and investigate the particle shell closing effects on the resulting cluster "magicity". This computational development can be used to establish rigorous structure−morphology−energetics relationships over the broadest range of material assemblies from thermodynamically small (and, hence, particularly challenging to describe with conventional thermodynamics) to nearly bulk-size. Our generalized Wulff construction technique can be developed further to also describe entropic as well as energetic effects and to allow for the creation of off-lattice structural features, such as stacking faults and other defective configurations.[15,16]

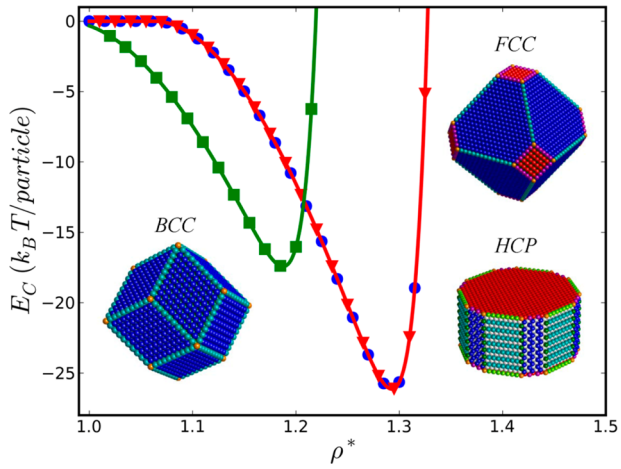

Figure 1. Cohesive energy of crystalline arrangements of AO colloidal particles. Cohesive energy, E_C, as a function of the reduced density, ρ^*, of the bulk phases of body-centered cubic (BCC), hexagonal close-packed (HCP), and face-centered cubic (FCC) AO colloidal crystals, shown as green, red, and blue solid curves, respectively. The inset configurations show the computed equilibrium shapes according to the Wulff construction for these three crystalline phases.

2. SIMULATION AND ANALYSIS METHODS

2.1. Interparticle Interaction Pair Potential and Energy Calculations. The pair potential governing the colloidal interactions used in this study, $U(r^*)$, has been shown to accurately model an experimental system containing colloidal silica particles in salt solution with poly(*N*-isopropylacrylamide) (PNIPAM) hydrogel particles.[17] While the functional form of this pair potential has been tuned to precisely describe the interactions in the above system, it can be easily adapted to describe interparticle interactions in any colloidal system characterized by a finite-ranged attraction with a short, relatively steep repulsion. $U(r^*)$ consists of two contributions and can be expressed as

$$\frac{U(r^*)}{k_B T} = \frac{U_E(r^*)}{k_B T} + \frac{U_D(r^*)}{k_B T} = \begin{cases} \dfrac{B}{k_B T}\exp(-\kappa^*(r^*-2)) - \Pi^*\left[\dfrac{4}{3}\pi(1+L^*)^3\left(1 - \dfrac{3r^*}{4(1+L^*)} + \dfrac{r^{*3}}{16(1+L^*)^3}\right)\right] & \text{if } r^* < 2 + 2L^* \\ 0 & \text{if } r^* \geq 2 + 2L^* \end{cases} \quad (1)$$

In eq 1, $r^* = r/a$, where r is the center-to-center interparticle distance and a is the particle radius which in this study is set at a = 1100 nm. B and κ^* are both parameters that control the electrostatic repulsion term, $U_E(r^*)$, in the pair potential and are set at $B/(k_B T)$ = 2497.0 and $\kappa^* = a\kappa \approx 140.5$, indicating a very short-ranged steep repulsion. Π^* and L^* control the depletant-based attractive term of the pair potential, where $\Pi^* = \Pi a^3/(k_B T)$ expresses the concentration of the PNIPAM depletant and affects the depth of the potential well and $L^* = L/a$ is a dimensionless parameter that describes the size of the PNIPAM depletant particles and affects both the well depth and the range of the interaction. The values of both Π^* and L^* are fixed in this study at Π^* = 146.41 and L^* = 0.1. All of the parameters in this pair potential, $B/(k_B T)$, κ^*, Π^*, and L^*, are chosen to match the experimental system described in refs 17 and 18. Previous work on this system has studied the thermodynamics and kinetics of small clusters of these colloidal particles over a relatively small size range.[19,20]

To calculate the energetic penalty associated with a given crystalline surface morphology consisting of facets with low-Miller-index crystallographic orientations and implement the conventional Wulff construction algorithm, we employ semi-infinite slabs with two free surfaces both of which have the same surface crystallographic orientation, characterized by the Miller indices $\{hkl\}$. Therefore, we can express the total slab energy, E_T, which is calculated from the pair potential, as the sum of the contributions to the total energy due to the presence of the free surfaces via the equation $E_T = NE_C + N_S\,\gamma_{\{hkl\}}$. Here, N is the total number of interacting colloidal particles in the slab, N_S is the total number of colloidal particles on both of the $\{hkl\}$-oriented free surfaces of the slab, E_C is the bulk cohesive energy of the crystal at its equilibrium density associated with the minimum of the corresponding cohesive energy curve of Figure 1, and $\gamma_{\{hkl\}}$ is the per-particle energetic penalty associated with this free surface; if γ were used to denote the (usual) per-surface-area energetic penalty associated with the formation of the free surfaces in the slab, then E_T could be expressed in an equivalent manner as $E_T = NE_C + 2\gamma A$, where A is the area of each free surface in the slab supercell. The total energy, E_T, of the slab is minimized following a conjugate-gradient algorithm as implemented in the open-source program LAMMPS,[21,22] with the energy and energy gradient computed according to the pair potential of eq 1. The surface energy of the $\{hkl\}$ surface, $\gamma_{\{hkl\}}$, is calculated as

$$\gamma_{\{hkl\}} = \frac{E_T - NE_C}{N_S} \quad (2)$$

Equation 2 is simply the result of solving the above equation for E_T for the only unknown quantity $\gamma_{\{hkl\}}$.

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2.2. Generalized Wulff Construction. Our implementation of the generalized Wulff construction consists of a set of lattice-site-exchange Monte Carlo (LSE-MC) simulations, which are described below, within a parallel tempering framework.[23] This technique is in some sense similar to the technique utilized by Northby to study clusters of Lennard-Jones atoms;[24] however, Northby's technique is carried out entirely at zero temperature and assumes a fixed cluster core structure, which significantly constrains the technique in terms of sampling configurational space. The parallel tempering framework is used to improve phase space sampling in the simulations and therefore reflect a more ergodic sampling of phase space than that achieved by standard Monte Carlo simulation; however, ergodicity is not guaranteed, and the predictions of our technique represent, strictly speaking, upper-bound estimates of the lowest-energy system states. Each generalized Wulff construction calculation results in the minimum-energy configuration for a given system size, N, and choice of underlying crystalline lattice structure (FCC, HCP, BCC, etc.). The initial configuration of a single LSE-MC simulation is a set of N particles placed within a set of M possible crystalline lattice sites such that $M > N$. The M crystalline lattice sites used in the LSE-MC simulation correspond to the particle positions of a crystalline cluster obtained by the Wulff construction for a specified crystal packing and containing ∼10 times more particles than the number of particles N, considered in the LSE-MC process. Additionally, we fix the coordinates of a single particle occupying the lattice site at the center of the cluster, which guarantees that the cluster center of mass is located far from the edge of the M-site cluster. These two conditions, namely, $M \sim 10N$ and the cluster center of mass constraint, ensure that the results of the generalized Wulff construction are not affected by the actual size of the M-site cluster obtained by the Wulff construction. A trial of the LSE-MC process attempts a swap of a particle to a new, and vacant, crystalline lattice site. Then, the entire resulting configuration is relaxed using 100 standard MC sweeps over all particles, except for the single particle fixed at the center of the set of lattice sites. Each sweep consists of an unbiased MC trial move for every particle, with the MC move accepted using geometrically increasing values of the effective inverse temperature, β, with $1/\beta$ ranging from $\sim 1.5 \times 10^{-4} k_B T$ to $0.5 k_B T$, effectively implementing a local relaxation scheme for this configuration. This set of trial moves, i.e., the particle lattice site swap and configurational relaxation, is accepted as a whole using the standard Metropolis criterion at an effective inverse temperature β_E. It should be noted that there are two inverse temperatures in this scheme, namely, β, the inverse temperature which is varied to perform the configurational relaxation step, and β_E, the inverse temperature which is used to determine whether an entire LSE move, the particle site swap and configurational relaxation together, are accepted. Several individual LSE-MC processes are run in parallel at different values of $1/\beta_E$ (generally ranging between $0.2 k_B T$ and $2 k_B T$). We attempt to switch the β_E values between two of these processes every 50 LSE-MC exchange moves, and this swap is accepted following the criterion described in ref 23. The temperature spacing is set, such that the temperature swap acceptance ratio is over the range from 0.60 to 0.75. The maximum temperature of $2 k_B T$ is set so that a move that breaks a single connection, which for this system requires an energy of $\sim 4.35 k_B T$, can be accepted with a reasonable probability, P (connection breaking move acceptance) ∼ 0.1.

2.3. Characterization of Minimum-Energy Configurations. To characterize the morphology and structure of the minimum-energy configurations of the small colloidal crystals generated, we consider the structure of the core around the center of the cluster. The core of the cluster is defined as the set of particles that are fully coordinated, i.e., those with $Z = 12$ for the close-packed crystal structures considered in this work. As a characterization tool, we calculate the distance from the center of mass of the set of fully coordinated particles to the nearest particle, D, and report the value of $D^* = D/(d_M)$, where d_M is the minimum-energy particle−particle separation, which for this system is $d_M \sim 2.06a$, with a being the particle radius. Normalization by this minimum energy separation, d_M, allows us to compare D^* values calculated from the fully connected core structures to those calculated from packings of hard spheres at contact as described below.[25,26] This dimensionless distance, D^*, provides a metric to describe the types of lattice sites around which the minimum-energy, equilibrium-shape crystal forms. Configurations that are centered on substitutional lattice sites, i.e., lattice points, have a single particle located at the center of mass of the fully connected core, and therefore $D^* = 0$.

Clusters with a $D^* \geq 0.5$ are centered at an interstitial (as opposed to substitutional) lattice site, which is not occupied by any individual colloidal particle. To characterize such sites, we consider shells of particles which are a constant distance away from a given type of interstitial site or void in the bulk FCC or HCP packing of hard spheres.[25] The D^* value of the configuration formed by considering every particle in the packing within a given shell around the void space centered at that interstitial site is the D^* value associated with that type of interstitial void. In our colloidal equilibrium crystals, we observe core structures which are packed around four different interstitial sites/voids: octahedral at $D^* = \sqrt{2}/2 = 0.7071$, tetrahedral at $D^* = \sqrt{6}/4 = 0.6123$, triangular at $D^* = 2\sqrt{3}/6 = 0.5774$ or $D^* = 0.6229$, and edge midpoint (or bond-centered) at $D^* = 0.5$ or 0.4582. The location of each of these sites in the FCC lattice and the local structure in their vicinity are shown in Figure 2, which depicts each of the interstitial sites connected to the particles occupying their nearest-neighboring lattice sites.[26] In the discussion of this characterization analysis, we often refer to closed-shell configurations around these interstitial sites or a lattice point: these are configurations that contain every particle from a given packing which is within a sphere centered on a particular interstitial site or on a lattice point. In Figure 2, the set of silver particles connected to an interstitial site create the smallest such shell around each of the individual interstitial sites. The triangular and bond-centered interstitial sites are associated with multiple D^* values: in the triangular interstitial case, the structure around the interstitial site grows anisotropically, which leads to differing values of D^* for different configurations grown around the same interstitial site. In the bond-centered interstitial case, the D^* value for a full shell around a bond-centered interstitial site is $D^* = 0.5$. However, in some configurations, the fully connected core is missing a single particle from what would otherwise be ten particles arranged in a closed shell around a bond-centered interstitial site, with the D^* value for this nine-particle configuration being $D^* = 0.4582$. While, technically, this different D^* value indicates that these configurations are not centered on a bond-centered interstitial site, we still consider them to be in the sense that they constitute configurations which have this single particle missing from a core that will, as

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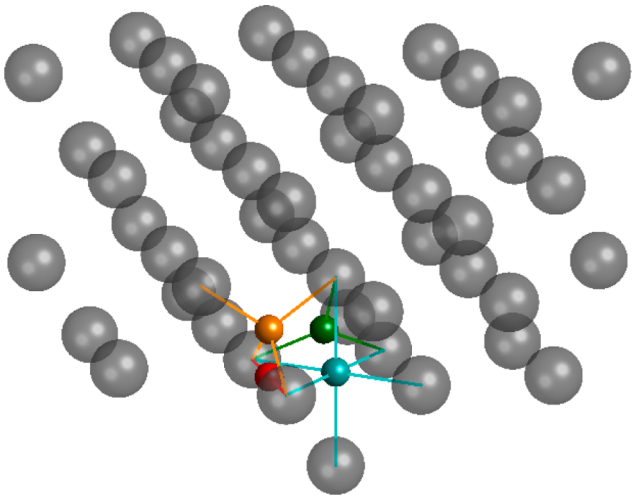

Figure 2. Illustration of interstitial sites used to classify minimum-energy colloidal cluster structure and morphology. The centers of the silver, translucent spheres correspond to lattice sites in the FCC crystalline lattice structure, while the centers of the smaller colored spheres denote the interstitial sites under consideration. The interstitial sites identified are the edge midpoint (or bond-centered), triangular, tetrahedral, and octahedral sites with the assigned spheres colored as red, green, orange, and cyan, respectively. The interstitial sites are connected to their nearest-neighboring lattice sites in the packing with line segments of the same color. The sizes of the spheres are chosen for visual clarity of the interstitial space of the lattice and the location of the interstitial sites. It should be noted that although this schematic depiction of colloidal crystal structure corresponds to the FCC lattice, all of the interstitial sites shown also are present in the HCP lattice.

the system size increases, become the ten-particle closed-shell core. In the results of the generalized Wulff construction several sites are observed at $D^* > 0.5$ but cannot be identified as any one of the types of high-symmetry interstitial sites listed above. These sites are, in general, intermediate sites between those listed above, and we refer to them as "low-symmetry" sites.

The sites with $0 < D^* < 0.5$ are located very close to a colloidal particle and are classified as either bilayer crystalline cluster centers or low-symmetry points. The bilayer crystalline cluster centers have two D^* values, 0.3537 or 0.3771; in all of the clusters sizes we have considered, this bilayer crystalline cluster consists of 12 or 13 particles arranged in two adjacent {111} layers. In the 13-particle bilayer cluster, the structure consists of a close-packed planar hexagonal arrangement of 7 particles (6 particles around a central particle) above a close-packed triangular arrangement of 6 particles with a $D^* = 0.3771$, while in the 12-particle bilayer cluster it consists of the same hexagonal arrangement above a triangular one, missing a single corner particle, with a $D^* = 0.3537$. Low-symmetry sites at the center of core regions with D^* values different from all of the above correspond to core structures which are either intermediate between those formed around the different types of interstitial and lattice points, or bilayer crystalline cluster structures, or have cores that are not fully formed, similar to the $D^* = 0.4582$ core around the bond-centered site or the $D^* = 0.3537$ bilayer crystalline clusters; however they are missing, in general, more than one particle or not consistently missing the same particle.

3. RESULTS AND DISCUSSION

The first step toward determining the minimum-energy configurations of such large and small colloidal crystals is the implementation of the conventional Wulff construction. This is an energy minimization calculation, where the energetic penalties associated with forming surface facets in a finite-sized crystalline assembly of interacting particles is minimized, thereby minimizing the assembly's total energy.[12,14] The required input quantities to apply this technique are the energetic penalties (surface energies) associated with different types of facets.[12] The shapes of the facets are determined by the choice of crystalline structure and are characterized by the Miller indices which define each facet within a given crystalline structure. To carry out the Wulff construction, we have used the Wulff construction algorithm as implemented in the Atomic Simulation Environment python package.[27] The surface energy penalties required as input to implement the algorithm are calculated using eq 2, and the equilibrium-shape configurations are generated by the algorithm for given volume and particle density or given number of particles. One drawback of the Wulff construction is that the generated configurations are assumed to have a single particle on a crystalline lattice site at the center of mass of the crystalline cluster, which may be an assumption that constrains the algorithm from finding the configuration corresponding to the global energy minimum for the given cluster size.

3.1. Equilibrium Shapes of Colloidal Crystals According to the Wulff Construction. Figure 1 shows a set of cohesive energy curves for the energy per particle of bulk colloidal crystals as a function of their reduced density, $\rho^* = N\sigma^3/L^3$, where N is the total number of particles in the simulation box, σ is the particle diameter, and L is the length of the simulation box edge, for three perfect crystalline packing arrangements, namely, face-centered cubic (FCC), hexagonal close-packed (HCP), and body-centered cubic (BCC). The short-ranged nature of the particle interactions, described in the Simulation and Analysis Methods section, results in no difference in cohesive energy, E_C, between the FCC and HCP crystals, as expected given that both packings have the same maximum particle coordination number, $Z = 12$. The BCC structure has a much higher energy as its particle coordination number is much lower, $Z = 8$, than that of the FCC and HCP structures.

The inset configurations in Figure 1 show the results of the Wulff construction carried out for the FCC, HCP, and BCC crystal structures for a relatively large number of particles, $N \sim O(10^3)$ to $O(10^4)$.[27] For the FCC crystal, the Wulff construction result shown is an assembly of $N = 12\,858$ particles with the shape of a truncated octahedron consisting of {100} and {111} facets; other low-Miller-index facets, including {110}, {210}, {211}, etc., were considered but did not affect the computed equilibrium morphology. For the HCP crystal, the Wulff construction result shown is an assembly of $N = 9915$ particles in a truncated hexagonal prism with {0001} basal-plane facets and $\{10\bar{1}0\}$ or $\{2\bar{1}\bar{1}0\}$ edge-plane facets. The Wulff construction shown for the BCC structure consists of 4641 particles forming a rhombic dodecahedron with only {110} facets. In all of these equilibrium-shape configurations, the surface particles are colored by the type of facet which they are on. To our knowledge, the equilibrium shapes of these bulk-sized colloidal crystals have not been reported in the literature.

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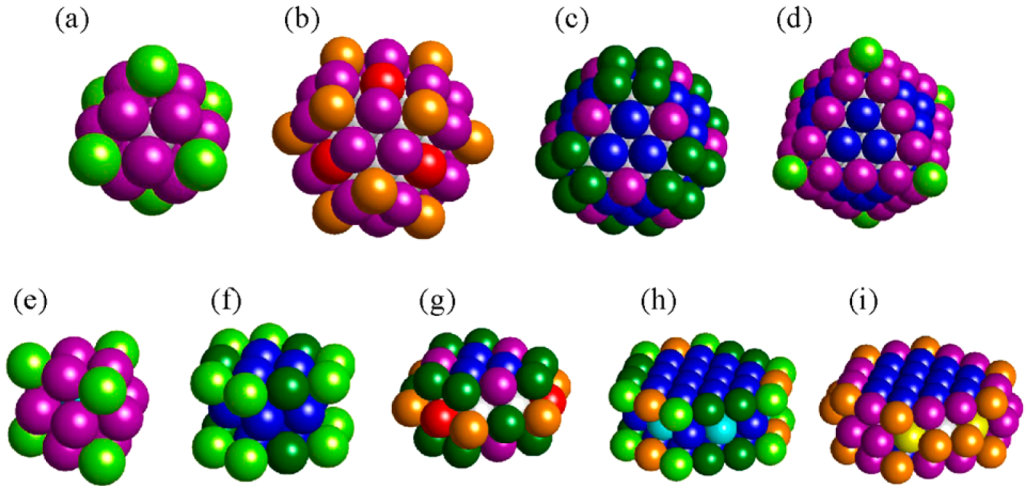

Figure 3. Minimum-energy configurations obtained by the Wulff construction. Equilibrium morphologies according to the Wulff construction of AO colloidal clusters of small size (number of particles) N with (a−d) FCC and (e−i) HCP crystalline lattice structure. (a) $N = 19$, (b) $N = 55$, (c) $N = 79$, (d) $N = 85$, (e) $N = 19$, (f) $N = 31$, (g) $N = 43$, (h) $N = 73$, and (i) $N = 85$. Each particle in the shown clusters is colored according to its coordination number (Z); light green, gold, green, magenta, red, blue, teal, yellow, and white colored particles have coordination number $Z = 4$, 5, 6, 7, 8, 9, 10, 11, and 12, respectively.

By identifying the individual particles as being either on a single facet, or on an edge between two or more facets, or as part of the crystalline bulk, we can express an estimate of the cluster energy taking only the terms accounted for by the Wulff construction. This energy estimate, E_W, is given by the equation

$$E_W = NE_C + \sum_{\{hkl\}} \gamma_{\{hkl\}} N_{\{hkl\}} \tag{3}$$

In eq 3, N is the total number of colloidal particles in the cluster, E_C is the cohesive energy of the bulk crystal, $\gamma_{\{hkl\}}$ is the per-particle energetic penalty associated with a facet of $\{hkl\}$ crystallographic orientation, which is calculated from eq 2 and is an input to the Wulff construction, and $N_{\{hkl\}}$ is the total number of particles identified to be on a given $\{hkl\}$ surface facet. It should be noted that care must be taken in assigning particles which are on an edge between two facets or on a vertex between three or more facets to avoid overcounting. In the estimates that follow, we have assigned these edge or vertex particles fractionally to each of the facets which they border (i.e., if a particle in the FCC crystal is on an edge between {100} and {111} facets, it is counted as 1/2 of a {100} particle and another 1/2 of a {111} particle). Because of the symmetry of the equilibrium shapes according to the Wulff construction, this fractional assignment does not result in fractional numbers of surface atoms in the final sum. The resulting estimate can be compared to the total energy of the crystal calculated using the energy according to the pair potential, E_P, from a conjugate-gradient energy minimization on the assembly generated by the Wulff construction. A good agreement between E_W and E_P, measured by $\Delta E = |E_W - E_P|$, indicates that the Wulff construction can capture all the relevant energetic effects on the cluster and that the shape generated by the Wulff construction is the actual equilibrium morphology for the given crystal packing and cluster size N. The FCC truncated octahedron of Figure 1 has an energy of $E_P = -24.709k_BT$/particle, which is in good agreement with E_W with $\Delta E = 0.01k_BT$/particle. The HCP assembly of Figure 1 has an energy of $E_P = -24.322k_BT$/particle, showing a similarly good agreement with $\Delta E = 0.06k_BT$/particle. Consistently with the close-packed crystals, the BCC assembly has an energy of $E_P = -16.181k_BT$/particle with $\Delta E = 0.11\ k_BT$/particle, also indicating that the Wulff construction is able to accurately describe minimum-energy configurations over the corresponding size range.

Representative equilibrium colloidal crystal shapes according to the Wulff construction carried out for configurations which are much smaller than the configurations shown as insets in Figure 1, with sizes corresponding to numbers of particles of $O(10)$ to $O(10^2)$, are shown in Figure 3 for the FCC and HCP crystal structures. We mention that from this point on we have chosen to focus on only the FCC and HCP crystalline structures since the BCC crystalline structure is characterized by a significantly smaller maximum number of connections between particles (8 compared to 12 for the FCC and HCP structures), which guarantees that the BCC structure is much less stable than the close-packed structures in the absence of extreme conditions, such as high stresses. In these configurations, the particles are colored according to their coordination number. For short-ranged pair potentials, the individual particle coordination number is a proxy for the energy that the particle contributes to the total cluster energy.[28] The shapes of both the FCC and HCP clusters over this small size range show strong size dependence. The morphology of the FCC clusters in Figures 3a and 3d, which contain $N = 19$ and $N = 85$ particles, respectively, is octahedral. The morphology of the cluster of Figure 3b, which contains $N = 55$ particles, is a cuboctahedron, while that of the cluster of Figure 3c, which contains $N = 79$ particles, is the same truncated octahedral shape as in the bulk case. The shape of the HCP cluster shown in Figure 3e, which contains $N = 19$ particles, is a triangular prism, while those of the HCP clusters in Figures 3f−i, which contain $N = 31$, $N = 43$, $N = 73$, and $N = 85$ particles, respectively, are all irregular hexagonal prisms. There are two HCP equilibrium-shape clusters in this size range with $N = 37$ and $N = 61$, which are not depicted in Figure 3, that both contain particles with very low coordination numbers ($Z = 3$ or 4) which are not part of any facet. It is important to note that for any crystal packing the Wulff construction cannot create crystals of arbitrary numbers of particles, as it is carried out for a fixed volume.

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While for the equilibrium-shape large clusters shown in Figure 1 the energy predicted via the Wulff construction estimate and the total energy calculated via the pair potential are in good agreement with each other (resulting in low values of $\Delta E = |E_W - E_P|$), the same agreement between these two determinations of the equilibrium crystal state is not observed when the crystal size is in the small-size regime. For all of the FCC clusters shown in Figures 3a–d, the energies according to the Wulff construction and the pair potential differ from each other by more than $1.5k_BT$/particle. For the HCP clusters shown in Figures 3e–i, the corresponding agreement between the two calculations is similarly poor. This difference in the computed energies indicates that the Wulff construction cannot account for the effects of the relatively high proportion of edge and vertex particles in the small crystal surface compared to particles both in the crystalline bulk and on single facets of the crystalline surface morphology.

3.2. Equilibrium Shape of Colloidal Crystals According to the Generalized Wulff Construction. Based on the above results, it is evident that the Wulff construction does not predict accurately the equilibrium-state, minimum-energy configuration of the small colloidal crystals. Consequently, when considering these types of small clusters, a generalized Wulff construction is required. Our implementation of such a generalized Wulff construction is based on a lattice-site-exchange Monte Carlo (LSE-MC) scheme within a parallel tempering framework, which is described in detail in the Simulation and Analysis Methods section. Representative results of the generalized Wulff construction, carried out for the 55-particle FCC cluster, are shown in Figure 4, which shows the progress of ten LSE-MC simulations carried out within a parallel tempering framework at β_E values equally spaced over the interval $0.2k_BT \leq 1/\beta_E \leq 2k_BT$. The individual LSE-MC energy evolution sequences are colored according to the corresponding β_E values. The dashed magenta line in Figure 4 marks the energy of the crystalline cluster generated by the Wulff construction, and the inset in Figure 4 shows two views of the minimum-energy configuration generated by this generalized Wulff approach; the energy of this crystal is lower than that of the crystal generated by the Wulff construction. The inset configuration in Figure 4 is clearly different than the highly symmetric crystal with cuboctahedral morphology generated by the Wulff construction and shown in Figure 3b.

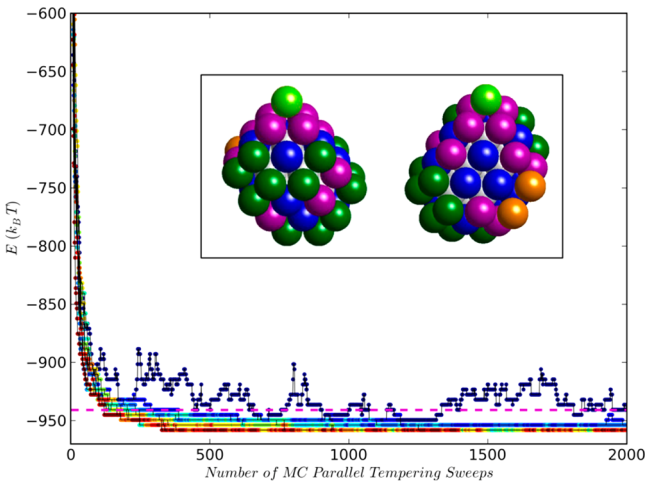

Figure 4. Energy evolution during the progress of the generalized Wulff construction. Energy evolution over the course of the generalized Wulff construction for the FCC colloidal cluster of N = 55; different energy evolution sequences correspond to different parallel tempering replicas of the 55-particle cluster. The magenta dashed line marks the energy of the 55-particle cluster generated according to the conventional Wulff construction. The inset shows two views of the minimum-energy cluster found for this cluster size. Particles are colored according to their coordination number following the convention of Figure 3.

Figure 5 shows the results, for equilibrium cluster energy E and morphology as a function of cluster size N, of the generalized Wulff construction over the size range $6 < N < 86$, carried out for both FCC and HCP clusters; $-E(N)$ is plotted, implying that more stable clusters have a higher $-E$. While the generalized Wulff construction can be carried out for larger clusters, with many more than ∼90 particles, we have limited this study to crystalline clusters with well under 100 particles, since the focus of this work is on enabling predictions of equilibrium crystal morphologies for thermodynamically small

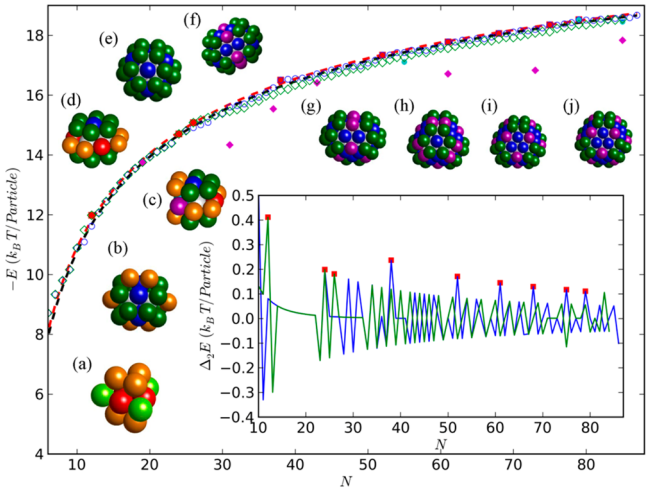

Figure 5. Energetics of equilibrium-shape small colloidal crystals. Energy, E, of colloidal clusters with equilibrium morphologies according to the generalized Wulff construction as a function of the cluster size, N, and its crystalline lattice structure, over the range $6 \leq N \leq 87$; blue open circles and green open diamonds denote energies of FCC and HCP clusters, respectively. The energies of FCC and HCP clusters generated according to the conventional Wulff construction are denoted by solid cyan circles and solid magenta diamonds, respectively. Red solid squares indicate "magic" cluster sizes. The two dashed lines correspond to two optimal fits of the functional form $|E(N)|/N = E_C + \hat{\gamma}N^{-1/3}$ to the shown $E(N)$ data, where E_C is the cohesive energy of the bulk crystal and $\hat{\gamma}$ is a fitting parameter representing an average energetic penalty associated with the formation of a single surface particle. Specifically, the black and red dashed lines correspond to a fit of the above functional form over all the cluster sizes examined and over only the magic cluster sizes, respectively. The plot in the inset gives as a function of N the second-order energy difference $\Delta_2E(N) = E(N+1) + E(N-1) - 2E(N)$ of the equilibrium configurations generated by the generalized Wulff construction for the FCC and HCP clusters shown as blue and green solid lines, respectively, for $10 \leq N \leq 87$; red solid squares are used to denote "magic" cluster sizes. (a–j) Minimum-energy cluster configurations for the identified magic cluster sizes: (a) FCC at N = 12, (b) FCC at N = 24, (c) HCP at N = 24, (d) HCP at N = 26, (e) FCC at N = 38, (f) FCC at N = 52, (g) FCC at N = 61, (h) FCC at N = 68, (i) FCC at N = 75, and (j) FCC at N = 79. The particles in these inset configurations are colored according to their coordination number following the convention of Figure 3.

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assemblies far removed from the near-bulk size scale. It should be mentioned that pursuing this study for clusters with well over 100 particles will identify the size limit beyond which the predictions of the traditional Wulff construction remain accurate. In Figure 5, the cluster energies according to the generalized Wulff construction are denoted as open blue circles and open green diamonds for the FCC and HCP crystal structures, respectively. For every cluster size $N \leq 30$, the minimum-energy configurations for both crystal packings have identical energies except for $N \in \{11, 12, 26, 27, 28\}$, where the crystal with HCP lattice structure is more stable. For clusters of $N \geq 31$, the resulting crystal of FCC lattice structure is more stable than that of HCP structure for every cluster size except for $N = 83$, for which the resulting FCC and HCP clusters have the same energy. The filled cyan circles and the filled magenta diamonds denote the energies of the configurations generated by the conventional Wulff construction for the FCC and HCP clusters, respectively. It is evident that at given N and lattice structure the crystals generated by the conventional Wulff construction are consistently less stable than those generated by the generalized Wulff construction. There are only two cases, over the entire size range examined, for which the crystalline configurations generated by both the Wulff and the generalized Wulff construction have the same energy: $N = 19$ for both crystalline packings, FCC and HCP, and $N = 79$ for the FCC lattice structure.

In order to characterize the relative stability of similarly sized clusters, we calculate the second-order energy difference, $\Delta_2 E(N) = E(N+1) + E(N-1) - 2E(N)$.[29] Clusters with high values of $\Delta_2 E(N)$ are considered to be "magic", as they are significantly more stable than clusters of comparable size.[13,29] The inset plot in Figure 5 shows $\Delta_2 E(N)$ with red filled squares used to denote the magic cluster sizes as is also done in the main $-E(N)$ plot of Figure 5. In Figures 5a–j, the equilibrium configurations are shown of the magic sized crystalline clusters generated by the generalized Wulff construction. The equilibrium crystals of Figures 5a,c,d at $N = 12$, 24, and 26 particles, respectively, are for an HCP crystalline packing, while those of Figures 5b,e–j at $N = 24$, 38, 52, 61, 68, 75, and 79 particles, respectively, are for an FCC lattice structure. The structure and morphology of these magic clusters are characterized in detail below.

The black and red dashed curves in Figure 5 are least-squares fits for the per particle energy as a function of the cluster size performed over the full set of cluster sizes and for only magic sized clusters, respectively. For this fitting procedure, we assumed for the size dependence of the equilibrium cluster energy the functional form $|E|/N = E_C + \hat{\gamma} N^{-1/3}$. This functional form is derived from eq 3 by making two assumptions: first, there exists a thermodynamic property $\hat{\gamma}$ which characterizes the average energetic penalty associated with the formation of a single surface atom and, second, for a given cluster size the number of particles on the surface of the cluster is proportional to $N^{2/3}$, an assumption equivalent to a surface area being proportional to the square of the cluster radius, an exact statement for an isotropic spherical cluster. The quantity $\hat{\gamma}$ is the only fitting parameter in this procedure with E_C taken equal to the cohesive energy of the bulk crystal. Fitting over the entire data set gives $\hat{\gamma} = 33.09 k_B T$, while fitting over the magic clusters only gives $\hat{\gamma} = 32.79 k_B T$. While the energetic penalty for the magic clusters fit is slightly lower, as expected, both of these values give a good description of the crystalline cluster energetics according to the generalized Wulff construction. When the same fitting procedure is applied to the results for the size dependence of the energy of clusters generated according to the Wulff construction, we find that the quality of the fit is lower (given the more polyhedral-like morphology of the Wulff-constructed clusters) and that $\hat{\gamma} = 35.02 k_B T$, i.e., a value higher than the corresponding values for the magic clusters or the entire set of the generalized Wulff constructed clusters, as expected since the clusters generated according to the standard Wulff construction are systematically less stable than those generated according to the generalized Wulff construction.

3.3. Hyperstatic Energy Description and Reductions of Energetic Degeneracy. A different description of the overall cluster energy can be constructed by calculating the total number of contacts in the cluster.[16] In Figure 6, we report this total number of contacts less the minimum number of contacts necessary to create a mechanically stable configuration[28]

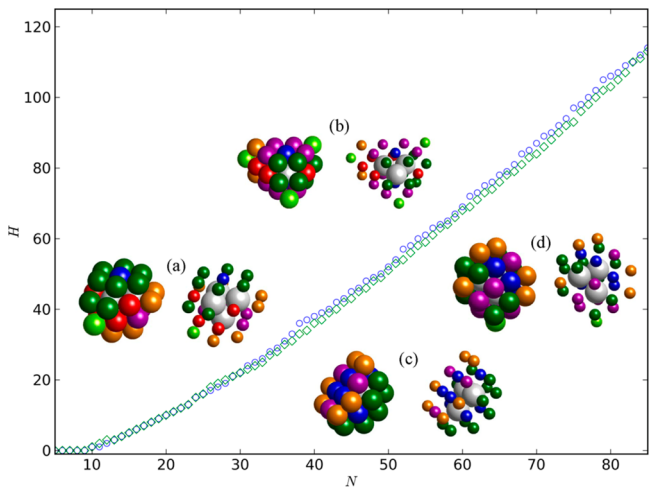

Figure 6. Hyperstaticity of equilibrium-shape small colloidal crystals. Hyperstaticity, H, defined as the total coordination number of a cluster reduced by $(3N - 6)$, the required total coordination number to guarantee a rigid molecule, for minimum-energy FCC and HCP crystalline clusters over the size range $5 \leq N \leq 85$; blue open circles and green open diamonds denote hyperstaticities of FCC and HCP clusters, respectively. The total coordination number serves as a proxy for the cluster energy for these types of short-ranged pair potentials. For cluster sizes $N \leq 30$, the total coordination numbers of the FCC and HCP clusters are exactly the same, except for the six cluster sizes $N \in \{5, 11, 12, 26, 27, 28\}$. At $N = 5$, the total coordination number of the FCC cluster is higher by 1 than that of the corresponding HCP cluster. For $N \in \{11, 12, 26, 27, 28\}$, the total coordination number of the HCP cluster is higher by 1 than that of the corresponding FCC cluster. For cluster sizes $31 \leq N \leq 85$, the FCC clusters are consistently characterized by a higher total coordination number than the HCP clusters of equal size, except for $N = 83$ where both the FCC and the HCP clusters have the same total coordination number. The FCC and HCP cluster configurations, together with their corresponding core configurations, are shown as insets for $N = 30$ and $N = 31$, the sizes beyond which the total coordination number of the minimum-energy FCC cluster surpasses that of the corresponding HCP cluster. In each pair of these inset configurations, the spheres representing surface particles in the configuration on the right have been reduced in size to allow for a clear visualization of the cluster's core structure. (a) HCP at $N = 30$, (b) HCP at $N = 31$, (c) FCC at $N = 30$, and (d) FCC at $N = 31$. The particles in these inset configurations are colored according to their coordination number following the convention of Figure 3.

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$$H = \sum_{i=1}^{N} \frac{Z_i}{2} - (3N - 6) \tag{4}$$

for the equilibrium-shape colloidal crystals generated by the generalized Wulff construction. In eq 4, Z_i is the coordination number of particle i, N is the total number of particles in the configuration, and H is termed the hyperstaticity of the cluster.[30] Recent studies indicate that increasing the hyperstaticity of a cluster leads to reduction of the observed energetic degeneracy not only of configurations with off-lattice structural features, such as stacking faults or low-coordination structural defects, but also of on-lattice configurations, such as perfect FCC- and HCP-based configurations.[15,30] This reduction in possible energetic degeneracy may explain why the FCC and HCP equilibrium-shape crystals become no longer energetically degenerate between $N = 30$ and $N = 31$. In addition to considering the hyperstaticity of the clusters, we also visualize the equilibrium-shape crystalline configurations at $N = 30$ and $N = 31$ corresponding to the FCC and HCP packings and shown as insets in Figures 6a–d. These crystalline cluster configurations highlight the changes in the structure and morphology of these minimum-energy configurations occurring as the cluster size N increases from 30 to 31.

3.4. Characterization of Equilibrium-Shape Colloidal Crystals According to the Generalized Wulff Construction. To characterize the morphology and structure of the minimum-energy crystalline cluster configurations, we consider the structure of the core around which each cluster is growing. In Figure 7, we report the dimensionless distance D^* (defined fully in the Simulation and Analysis Methods section) which is proportional to the actual distance from the center of mass of the core of maximally coordinated particles of the minimum-energy configuration to the nearest neighboring particle. We have identified six types of interstitial sites on which these minimum-energy configurations are centered: edge midpoint or bond-centered, triangular, tetrahedral, and octahedral interstitial sites, bilayer crystalline cluster centers, and low-symmetry interstitial sites. In Figure 7, the plots on the left and on the right show the D^* values for the equilibrium-shape crystals according to the generalized Wulff construction with FCC and HCP packings, respectively. For the very small clusters with $3 \leq N \leq 13$, both the FCC and HCP crystals form the smallest possible structures around interstitial sites, such as four particles located in the vertices of a tetrahedron around the interstitial site or six particles occupying the vertices of an octahedron around the interstitial site, as well as intermediate sites between these high-symmetry interstitial sites. It should be noted that for configurations in this very small size range there is no fully connected core, and therefore, the D^* value is calculated over the entire cluster. In this size range, there is a single magic cluster with HCP packing at $N = 12$, which is a closed-shell configuration centered on a triangular interstitial site.

The first cluster with a fully connected core particle forms at $N = 14$. From this size through $N = 28$, the minimum-energy clusters for crystals with both FCC and HCP lattice structure transition over similar core structures. In this size range, there are two magic clusters at $N = 24$ and $N = 26$, with cores forming around bond-centered and triangular interstitials, respectively. The cluster at $N = 24$ is a closed-shell one for both the FCC and HCP packings. At $N = 26$, only the cluster with HCP packing has a closed-shell structure at this cluster size, which explains the improved stability of HCP crystalline clusters of $N = 26$–28.

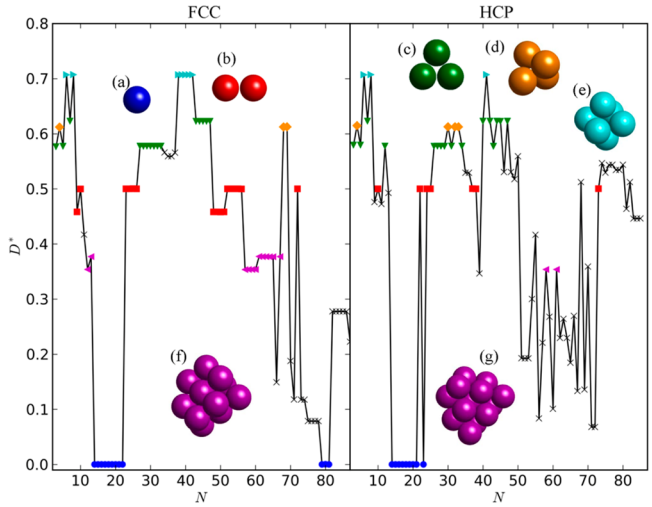

Figure 7. Structural and morphological characterization of equilibrium-shape small colloidal crystals. Structural and morphological characterization of equilibrium-shape colloidal clusters according to the generalized Wulff construction over the size range $3 \leq N \leq 87$. The dimensionless distance D^* from the center of mass of the fully coordinated core (i.e., the set of 12-fold coordinated particles) to the nearest neighbor is plotted as a function of cluster size N and crystalline lattice structure (FCC or HCP). Clusters are characterized by the type of lattice site located at the center of the cluster core: blue circles, green triangles, red squares, magenta triangles, orange diamonds, teal triangles, and black crosses are used to denote lattice points, triangular interstitial sites, bond-centered interstitial sites, bilayer crystalline cluster centers, tetrahedral interstitial sites, octahedral interstitial sites, and low-symmetry interstitial sites, respectively. The inset configurations show representative core structures centered at a (a) crystal lattice point, (b) bond-centered interstitial site, (c) triangular interstitial site, (d) tetrahedral interstitial site, (e) octahedral interstitial site, and (f, g) bilayer crystalline cluster center. The particles in these inset configurations are colored consistently with the color code of the symbols used in the plot to denote the lattice site that they represent.

For $N > 28$, the cluster equilibrium morphologies of the FCC and HCP packings differ. In FCC crystals, we observe relatively stable core configurations with transitions between well-defined structures and few configurations centered on low-symmetry sites until $N \geq 70$. The HCP crystals exhibit a different behavior, with multiple transitions between core structures and a large number of low-symmetry sites, especially for clusters of more than 46 particles. The magic cluster sizes identified from the generalized Wulff construction are characterized by cores centered on an octahedral interstitial site, bond-centered interstitial site, bilayer crystalline cluster center, tetrahedral ineterstitial site, and a substitutional lattice site at $N = 38$, 52, 61, 68, and 79, respectively. The only magic cluster size with a core centered on a low-symmetry site is the cluster at $N = 75$: the core of this magic cluster consists of 18 particles arranged in an octahedron missing a single vertex particle. While this configuration does not form a closed shell in either its core or its outer structure, it constitutes a certain intermediate configuration that leads to the magic cluster of $N = 79$, which contains the full 19-particle closed-shell octahedron as its core.

The equilibrium clusters centered on a substitutional lattice site, over the ranges $14 \leq N \leq 21$ in HCP and FCC packings

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and $79 \leq N \leq 81$ in the FCC packing, contain two configurations of particular interest, the clusters at $N = 19$ and $N = 79$ particles. These clusters are the only two where the configurations from the Wulff construction have identical energy with those generated according to the generalized Wulff construction. This agreement is expected, since the configurations generated by the Wulff construction are based on a lattice which has a single particle centered at a lattice point at its origin. The fact that the majority of the minimum-energy configurations generated by the generalized Wulff construction are not centered on lattice points reinforces the conclusion that the generalized approach is necessary to accurately determine the minimum-energy, equilibrium shape configuration.

While many of these magic clusters consist of closed shells formed around an interstitial site or lattice point, such as the FCC octahedral cluster at $N = 38$ or FCC and HCP clusters forming around a bond-centered site at $N = 24$, it is not only the closure of the outer-shell structure that dictates the cluster morphology or whether a given cluster is magic. For example, the FCC cluster forming around a bond-centered site at $N = 52$ is not a closed-shell structure around its high-symmetry central site. Instead, the FCC cluster at $N = 52$ is the first one with a 10-particle fully connected core. This core corresponds to a closed-shell configuration around a bond-centered interstitial site, and it is this closure of the fully connected core which leads to the improved stability of the $N = 52$ particle cluster, not the outer-shell closure which occurs at $N = 56$.[25] Additionally, the FCC magic cluster at $N = 68$ is centered on a tetrahedral site; however, this is not the only possible closed-shell morphology for a 68-particle FCC cluster. In the FCC packing, a closed-shell configuration of 68 particles can be formed centered on either octahedral or tetrahedral interstitial sites.[25] The inner, fully connected core of the configuration centered on an octahedral interstitial site contains only 14 particles, while the core of the configuration centered on a tetrahedral interstitial site contains 16 particles, leading to a lower overall energy for the latter configuration. This interplay between the outer and inner shell closure leads to a particularly complex and hard to predict behavior for this cluster, which our method has been able to capture and characterize.

4. SUMMARY AND CONCLUSIONS

We have applied the Wulff construction to describe the equilibrium shape of colloidal clusters of a near-bulk size of $\mathcal{O}(10^3)$ to $\mathcal{O}(10^4)$ particles. While we are confident in the results of this technique for sizes over the above range and larger, we find that this conventional technique of computing equilibrium crystal shape cannot give an accurate description of the clusters' minimum-energy configurations when we apply it to small clusters of $\mathcal{O}(10)$ particles. To determine the minimum-energy configuration, and therefore the equilibrium shape of such small clusters, over the size range of $3 \leq N \leq 85$ particles, we have developed a generalized Wulff construction which consists of lattice-site-exchange Monte Carlo simulations within a parallel tempering framework. This technique does not rely on the same set of assumptions as the conventional Wulff construction that caused the conventional method to fail in the small cluster size range. The minimum-energy configurations resulting from the generalized Wulff construction for crystalline colloidal clusters with FCC and HCP lattice structures have per particle energies that exhibit a power law of $N^{-1/3}$ size dependence, with several configurations exhibiting "magicity", i.e., an improved stability compared to that of similarly sized clusters. The improved energetic stability of these magic clusters may play a significant kinetic role in the nucleation and growth of colloidal crystals. We have characterized in detail the morphology of the clusters generated by the generalized Wulff construction, which constitutes the equilibrium shape of these crystals, and related cluster core and shell closure effects to explain the existence of these magic clusters.

The equilibrium shape determination technique introduced in this study is not limited to colloidal crystals only or to colloidal clusters consisting of particles that interact through a certain force field. This is a general approach for minimum-energy and equilibrium shape determination of any assembly of atoms, molecules, or other particles interacting with each other through a known interaction potential or one that can be derived from first principles. The lattice-site-exchange Monte Carlo parallel tempering framework on which our method is based also can be extended further to include entropic effects, explicitly allow for polymorphic transitions between different lattice structures, and to account for lattice defects and structural disorder. Specifically, the parallel tempering technique can be generalized to allow not only for MC swaps with varying temperature, as is used in this study, but also for MC swaps between different crystalline lattice structures or defective states corresponding to various crystalline phase imperfections (such as stacking faults that can be generated by lattice plane displacements). This hyperparallel tempering type of MC scheme would naturally allow for the determination of the free energy differences between different structures and defects. Additionally, extending this technique to material systems with increased complexity, arising from large system sizes, existence of various structural features with different size scales, more complex morphology such as in multicentered molecular systems, or much stronger interatomic or interparticle interactions, will certainly face difficulties in terms of sampling and convergence. However, assuming adequately accurate interatomic or interparticle interaction potentials, careful tuning of the parameters of the parallel tempering technique will be able to handle these types of complex systems in a similar fashion to that analyzed in this study. In conjunction with the development in this study, such extensions will set the stage toward relating these minimum-energy configurations to the thermodynamics and kinetics of crystal nucleation and growth. This fundamental understanding of crystal nucleation and subsequent growth has far reaching implications for the development and engineering of photonic crystals and other "designer" materials or metamaterials, the constituents of which can be synthesized, interact with each other in a prescribed manner, and can be arranged in various architectures enabling new technologies. In addition to an improved understanding of crystalline phase nucleation in colloidal systems, understanding the stability of magic sized clusters allows for their use as individual units in photonic materials: these individual clusters of colloidal particles may form the constituent elements of a crystalline lattice, i.e., a regular array of particularly stable colloidal clusters. This type of "cluster crystal" can be engineered to have tunable material properties which can optimize its function as a photonic crystal. Moreover, we believe that these findings will motivate new experimental efforts on colloidal crystals and related materials and provide experimentally testable hypotheses to design and guide these experimental efforts.

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The authors declare no competing financial interest.

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